Prayas JEE AIR (2025)

Weekly Test 29

Advanced

DURATION : 180 Minutes | **DATE : 11/01/2025** | **M.MARKS : 180**

Topics Covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 51 questions. The maximum marks are **180**.

SECTION-1 (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : –1 In all other cases.

SECTION-2 (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 ONLY if (all) the correct option(s) is(are) chosen;

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : –2 In all other cases.

- For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then
  choosing ONLY (A), (B) and (D) will get +4 marks;
  choosing ONLY (A) and (B) will get +2 marks;
  choosing ONLY (A) and (D) will get +2 marks;
  choosing ONLY (B) and (D) will get +2 marks;
  choosing ONLY (A) will get +1 mark;
  choosing ONLY (B) will get +1 mark;
  choosing ONLY (D) will get +1 mark;
  choosing no option (i.e. the question is unanswered) will get 0 marks; and
  choosing any other combination of options will get –2 marks.

SECTION-3 (Maximum Marks: 24)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen
- virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If **ONLY** the correct integer is entered;

Zero Marks : 0 In all other cases.

SECTION-4 (Maximum Marks: 12)

This section contains **TWO (02)** paragraphs.

- Based on each paragraph, there are **TWO (02)** questions.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered in the designated place;

Zero Marks : 0 In all other cases.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ Invigilator's Signature _____

IMPORTANT CONSTANTS

Speed of light in free space,	:	$3.00 \times 10^{8}\ ms^{-1}$
Permeability of free space,	:	$4\pi \times 10^{-7}\ Hm^{-1}$
Permittivity of free space,	:	$8.85 \times 10^{-12}\ Fm^{-1}$
The Planck constant,	:	$6.63 \times 10^{-34}\ Js$
Rest mass of electron,	:	$9.1 \times 10^{-31}\ kg$
Rest mass of proton,	:	$1.67 \times 10^{-27}\ kg$
Molar gas constant,	:	$8.31\ JK^{-1}\ mol^{-1}$
The Avogadro constant,	:	$6.02 \times 10^{23}\ mol^{-1}$
The Boltzmann constant,	:	$1.38 \times 10^{-23}\ JK^{-1}$
Gravitational constant,	:	$6.67 \times 10^{-11}\ N\ m^2kg^{-2}$
Acceleration of free fall	:	$9.8\ ms^{-2}$
Rydberg Constant	:	$1.097 \times 10^{7}\ m^{-1}$
Atomic mass unit	:	$1.67 \times 10^{-27}\ kg$
Charge on proton	:	$1.6 \times 10^{-19}\ C$

IMPORTANT VALUES

$\sqrt{2}$ = 1.414

$\sqrt{3}$ = 1.732

$\sqrt{5}$ = 2.236

π = 3.142

e (Euler's constant) = 2.718

ln 10 = 2.303

$\log_{10}2$ = 0.3010

$\log_{10}3$= 0.4770

$\log_{10}7$ = 0.845

**** Use above values unless otherwise specified in a question.***

❑❑❑

PART-I (PHYSICS)

SECTION-1

Single Correct Type Questions

1. An ideal gas has molar heat capacity C_v at constant volume. The gas undergo the process $T = T_0 (1 - \alpha V)$ where α, T_0 is constant and V is volume. Then molar heat capacity of the gas is :

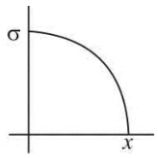
(A) $C_V + \frac{R}{\alpha V}(1-\alpha V)$

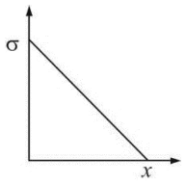
(B) $C_V - \frac{R}{\alpha V}(1-\alpha V)$

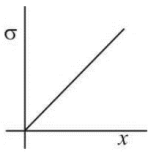
(C) $C_V - \frac{R}{2\alpha V}(1+\alpha V)$

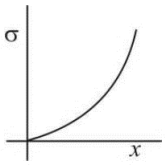
(D) $C_V + \frac{R}{2\alpha V}(1+\alpha V)$

2. A uniform rod rotating in gravity free region with certain constant angular velocity. The variation of tensile stress with distance x from axis of rotation is best represented by which of the following graphs:

(A) 

(B) 

(C) 

(D) 

3. The trajectory of light ray in a non-homogeneous medium is representing by $x = A\sin\left(\frac{y}{B}\right)$ where A and B are positive constants. Assuming that refractive index varies only with x-coordinate and has the value n_0 at $x = 0$; compute the refractive index as function of x in the space between the planes $x = -A$ and $x = +A$.

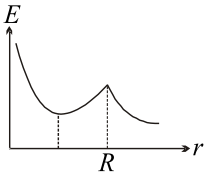
(A) $n(x) = n_0\left[1-\frac{x^2}{A^2+B^2}\right]^{1/2}$

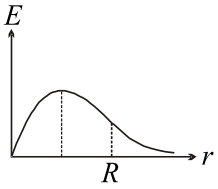
(B) $n(x) = n_0\left(1-\frac{x^2}{A^2}\right)^{1/2}$

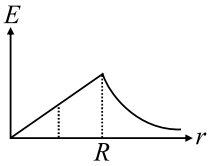
(C) $n(x) = n_0\left(1-\frac{x^2}{B^2}\right)^{1/2}$

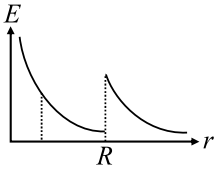
(D) $n(x) = \frac{n_0}{\left[1-\frac{x^2}{B^2}\right]^{1/2}}$

4. A spherical insulator of radius R is charged uniformly with a charge Q uniformly distributed throughout its volume and contains a point charge $\frac{Q}{16}$ located at its centre. Which of the following graphs best represent qualitatively, the variation of electric field intensity E with distance r from the centre.

(A) 

(B) 

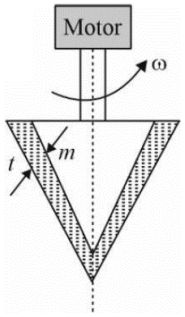
(C) 

(D) 

SECTION-2

One or More Than One Correct Type Questions

5. In the figure shown, there is a conical shaft rotating on a bearing of very small clearance t. The space between the conical shaft and the bearing, is filled with a viscous fluid having coefficient of viscosity η. The shaft is having radius R and height h. If the external torque applied by the motor is τ and the power delivered by the motor is P working in 100% efficiency to rotate the shaft with constant ω. Then



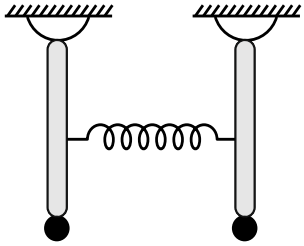
(A) $P = \frac{\pi\omega^2\eta R^3\sqrt{R^2+h^2}}{2t}$

(B) $\tau = \frac{\pi\omega\eta R^3\sqrt{R^2+h^2}}{2t}$

(C) $P = \frac{\pi^2\omega^2\eta R^3\sqrt{R^2+h^2}}{4t}$

(D) $\tau = \frac{\pi\omega\eta R^3\sqrt{R^2+h^2}}{4t}$

6. Some oscillating systems, instead of having one natural frequency, have two natural frequencies. Let the lower angular frequency be denoted by ω_L and the other one by ω_H. Consider one such system: Two physical pendulums, each comprising of a uniform rod of mass M and length L with small ball of mass M attached at one end, connected by a spring of spring constant k. The spring connects the mid-points of the rods and both the rods are hinged at their top points such that the whole arrangement is free to rotate in the vertical plane:



One of the natural frequencies results when both the rods are disturbed (together) by equal amounts in the same direction while the other natural frequency results when both the rods are disturbed (together) by equal amounts in opposite directions.

Now, consider the following statements and choose the correct option(s).

(A) Changing the stiffness of spring changes ω_H

(B) The ratio $\frac{\omega_H}{\omega_L}$ increases on increasing the dimension (L) of the system

(C) $\omega_L = \sqrt{\frac{9}{8}.\frac{g}{L}}$

(D) $\omega_H = \sqrt{\frac{3}{8}\left[\frac{2k}{M}+\frac{3g}{L}\right]}$

7. Consider a hypothetical situation in which the potential energy between electron and proton at a distance r is given by $\frac{-ke^2}{3r^3}$ where k is constant, e is charge of proton and r is radius of orbit. Application of Bohr's theory to hydrogen atom in this case shows that

(A) energy of electron in n^{th} orbit is proportional to n^6.

(B) energy of electron in n^{th} orbit is proportional to m^{-3} (m = mass of electron).

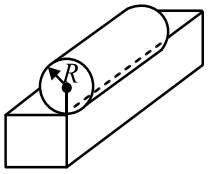
(C) energy of electron in n^{th} orbit is proportional to n^{-2}.

(D) energy of electron in n^{th} orbit is proportional to m^3 (m = mass of electron).

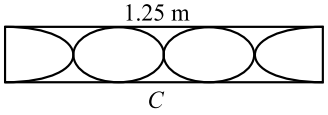
SECTION-3

Non-Negative Integer Type Questions

8. A rectangular rigid fixed block has a long horizontal edge. A solid homogeneous cylinder of radius R is placed horizontal ly at rest with its length parallel to the edge such that the axis of the cylinder and the edge of the block are in the same vertical plane as shown in Figure. There is sufficient friction present at the edge so that a very small displacement causes the cylinder to roll off the edge without slipping. The angle θ_c through which the cylinder rotates before it leaves contact with the edge is $\cos\theta_c = 4/x$. Find the value of x.



9. Two antinodes are observed on either side of rod C. If the frequency of the mode in which the rod is vibrating is f (Hz), find f /1000 ($Y = 2 \times 10^{11}$ N/m^2, density of rod = 8×10^3 kg/m^3, length of the rod = 1.25 m)

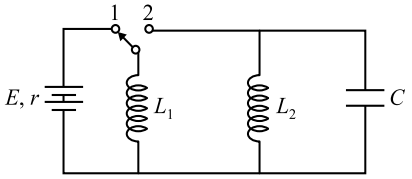


10. Two infinitely long straight wires are placed perpendicularly at a separation of d. Each wire carries linear charge density λ. The magnitude of repulsive force on each wire is $\frac{3\lambda^2}{n\varepsilon_0}$. Find the value of n.

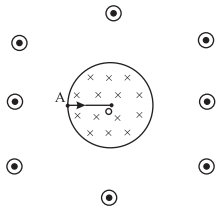
11. The radio isotope $^{64}_{29}Cu$ can decay through positron emission and electron capture. In both process neutrino is emitted. The difference in maximum possible kinetic energies of neutrino is k MeV. (given that rest mass energy of an electron is 0.51 MeV). Then the value of 100 k is?

12. A circuit shown consists of two inductors of inductances L_1 and L_2, a capacitor of capacitance C, a battery of electromotive force E and internal resistance r and a switch. Initially the switch was in position 1 for a long time. Find the maximum current (in A) in the inductor L_2 after the switch is thrown to position 2?

(Given: $E = 5V$, $L_1 = 4$ mH, $L_2 = 1$ mH, $r = 2\Omega$)



13. There exist a uniform inward magnetic field $B = 1\text{T}$ in cylindrical region of radius $r = 2\text{ m}$. Outside the cylindrical region the magnetic field is outward but of same magnitude 1 T. A point charge $(q = 1\text{C}$ and $m = 1\text{ kg})$ is projected from a point A on circumference with velocity 2 m/s towards center as shown in figure. After time $t = N\pi$ sec particle again passes through initial point A. Find the minimum possible value of *N*.

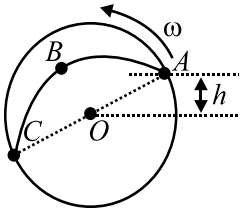


SECTION-4

Paragraph Type Questions

Paragraph for question nos. (14-15)

There is a hollow cylinder of radius *R* and it is rotating with constant angular speed ω about a horizontal axis passing through its center. There is a particle mass on circumference which rotates along with the cylinder and gets carried upward. Friction is sufficient enough such that the particle mass does not slip with respect to the cylinder as long as the normal force becomes zero. If the path of the particle after it lost contact with the cylinder is as shown along the curve *ABC* in the figure then,

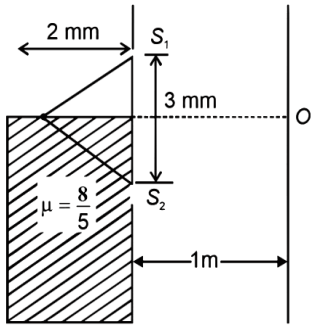


14. If the particle mass loses the contact at point *A* and follows trajectory *ABC* then the value of height *h* will be equal to $\frac{R\sqrt{x}}{2}$. Value of *x* is equal to _____. (Answer to nearest integer)

15. The value of ω (in terms of acceleration due to gravity *g* and *R*) is equal to $\sqrt{\frac{g\sqrt{y}}{4R}}$. Value of *y* is equal to _____. (Answer to nearest integer)

Paragraph for question nos. (16-17)

In a Young's double slit experiment, a point source is placed on a solid slab of refractive index 8/5 at a distance of 2 mm from the plane of two slits spaced 3 mm apart as shown and at equal distance from both the slits. The screen is at a distance of 1 m from the slits. Wavelength of light used is 500 nm in air.



16. Position of central maxima will be $\frac{1}{\sqrt{x}}$m below *O* then the value of *x* is.

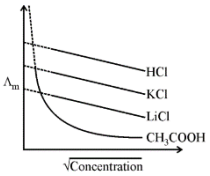
17. Order of fringe formed at *O* is 100*y*. Then the Value of *y* is _____.

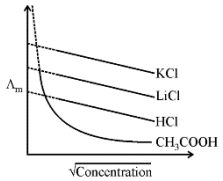
PART-II (CHEMISTRY)

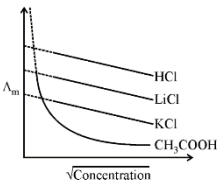
SECTION-1

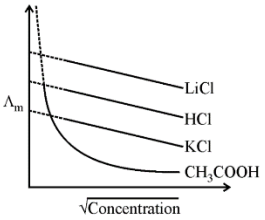
Single Correct Type Questions

18. Select the correct option for following aqueous solutions:

(A)


(B)


(C)


(D)


19. $H_3C - C \equiv C - Ph \xrightarrow{Hg^{2+}/H^+/H_2O} P$, P is:

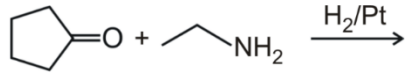
(A) $H_3C - \underset{\underset{O}{\|}}{C} - CH_2 - Ph$

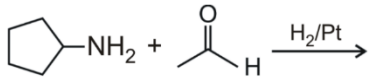
(B) $H_3C - CH_2 - \underset{\underset{O}{\|}}{C} - Ph$

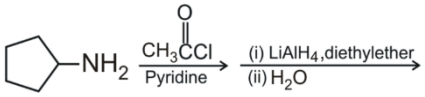
(C) $H_3C - \underset{\underset{O}{\|}}{C} - \underset{\underset{O}{\|}}{C} - Ph$

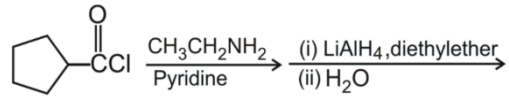
(D) $H_3C - \underset{\underset{OH}{|}}{CH} - CH_2 - Ph$

20. Among the following, which reaction will not give N-ethylcyclopentylamine as major product?

(A) 

(B) 

(C) 

(D) 

21. Potential energy of electron present in 2nd orbit of Li^{2+} is : (a_0 = Radius of 1st Bohr's orbit) :

(A) $-\frac{e^2}{4\pi\varepsilon_0 a_0}$ (B) $-\frac{3e^2}{4\pi\varepsilon_0 a_0}$

(C) $-\frac{3e^2}{16\pi\varepsilon_0 a_0}$ (D) $-\frac{9e^2}{16\pi\varepsilon_0 a_0}$

SECTION-2

One or More Than One Correct Type Questions

22. Consider two liquids A and B having pure vapour pressures P_A° & P_B° forming an ideal solution. The plot of $\frac{1}{X_A}$ y – axis v/s $\frac{1}{Y_A}$ x-axis (where X_A and Y_A are the mole fraction of liquid A in liquid and vapour phase respectively) is linear with slope and Y intercepts respectively:

(A) Intercept $= \frac{P_A^\circ - P_B^\circ}{P_B^\circ}$

(B) Intercept $= \frac{P_B^\circ - P_A^\circ}{P_B^\circ}$

(C) Slope $= \frac{P_A^\circ}{P_B^\circ}$

(D) Slope $= \frac{P_B^\circ}{P_A^\circ}$

23. The complex(es) which are paramagnetic in nature is/are:

(A) $[Cu(H_2O)_6]^{2+}$

(B) $[FeF_6]^{3-}$

(C) $[MnF_6]^{3-}$

(D) $[Fe(CN)_6]^{4-}$

24. First ionization energy order is CORRECT in:

(A) $Sc^{3+} > Sc^{2+} > Sc^{+}$

(B) $Sc^{3+} > Ti^{4+} > V^{5+}$

(C) Sc > Y > La

(D) Sc > Ca > K

SECTION-3

Non-Negative Integer Type Questions

25. Identify correct number of statements :

Statement-1: Entropy of system increases in dissociation of ammonia gas into N_2 and H_2 gases.

Statement-2: Entropy of system increases in polymerization process of ethene to form polythene.

Statement-3: Entropy of system increases in adiabatic irreversible compression of an ideal gas.

Statement-4: Entropy of system increases in condensation of H_2O (vapour).

Statement-5: Entropy of system increases in isothermal free expansion of ideal gas.

Statement-6: Entropy of system always increases in a spontaneous process.

Statement-7: Adiabatic process whether reversible or irreversible will not cause any change in entropy of the surroundings.

Statement-8: At absolute zero, entropy of any substance becomes zero.

26. 1.2 gram of carbon is burnt completely with 2.56 gram of oxygen to produce CO(g) and CO_2(g). None of the reactant remaining after reaction. Produced mixture of gases is treated with solid I_2O_5 (to know the amount of CO(g) produced). The liberated iodine required x mL of 0.1 M hypo solution for complete titration. The value of x is

27. Consider all possible isomeric ketones including stereoisomers of MW = 86 g/mol, All these isomers are independently reacted with $NaBH_4$ (NOTE: stereoisomers are also reacted separately). The total number of ketones that give a racemic product(s) is/are.

28. At 300 K temperature and 8 atm pressure the compressibility factor for a real gas is 0.8. The volume (in litre) of 10 mole of the gas at same temperature and pressure is? (Round off to the nearest integer)
[Given : R = 0.0821 L-atm mol^{-1} K^{-1}]

29. How many of the following will give N_2(g) on heating.
$(NH_4)_2SO_4$, NH_4NO_3, $(NH_4)_2Cr_2O_7$, NH_4Cl, NH_4NO_2, NH_4MnO_4.

30. Find the total stereoisomers for:

(a) $[Cr(C_2O_4)_3]^{3-}$ = x

(b) $[PtCl_2(en)_2]^{2+}$ = y

(c) $[Cr(NH_3)_2Cl_2(en)]^{+}$ = z

Hence write the value of (x + y + z).

SECTION-4

Paragraph Type Questions

Paragraph for question nos. (31-32)

For the reaction, $CHCl_3(g) + Cl_2(g) \xrightarrow{hv} CCl_4(g) + HCl(g)$ mechanism is given below

$$Cl_2(g) \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} 2Cl(g)$$

$$CHCl_3(g) + Cl(g) \xrightarrow{k_2} CCl_3(g) + HCl(g)$$

$$CCl_3(g) + Cl(g) \xrightarrow{k_3} CCl_4(g)$$

Given:

$k_1 = 4.8\times10^3\ sec^{-1}$, $k_{-1} = 1.2\times10^3\ M^{-1}sec^{-1}$,

$k_2 = 1.3\times10^{-2}\ M^{-1}sec^{-1}, k_3 = 2.1\times10^2\ M^{-1}sec^{-1}$

$E_{a_1} = 10\ kJ/mol$, $E_{a_{-1}} = 15\ kJ/mol$,

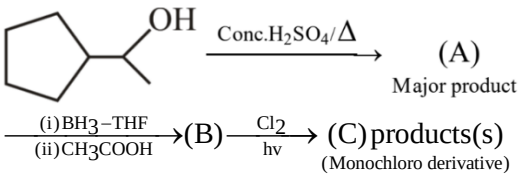
$E_{a_2} = 45 kJ/mol, E_{a_3} = 12\ kJ/mol$

31. Overall rate constant of reaction,

$CHCl_3(g) + Cl_2(g) \xrightarrow{hv} CCl_4(g) + HCl(g)$, is ____$\times10^{-2}\ M^{-1/2}\ sec^{-1}$.

32. Overall activation energy of reaction, $CHCl_3(g) + Cl_2(g) \xrightarrow{hv} CCl_4(g) + HCl(g)$, is ___ kJ/mol.

Paragraph for question nos. (33-34)



33. Total number of fractions obtained on fractional distillation of all monochloro derivatives product(s) (C) is _____?

34. The mass of Cl_2 required to react with 9.8 gm of compound (B) to form monochloro product is ___ grams. Assuming % yield = 100%
(Molar mass of Cl = 35.5 g/mol)

PART-III (MATHEMATICS)

SECTION-1

Single Correct Type Questions

35. The complex number z satisfies $|2z + 10 + 10i| \le 5\sqrt{3} - 5$, then the least principal argument of z is

(A) $\frac{-5\pi}{6}$ (B) $\frac{-11\pi}{12}$

(C) $\frac{-3\pi}{4}$ (D) $\frac{-2\pi}{3}$

36. The value of $\lim\limits_{x\to\infty}\left(\frac{x^2\sin\left(\frac{1}{x}\right) - x}{|x|}\right)$ is equal to

(A) 0 (B) 1

(C) 2 (D) 3

37. Radius of the circle that passes through origin and touches the parabola $y^2 = 4ax$ at the point $(a, 2a)$ is

(A) $\frac{5}{\sqrt{2}}a$ (B) $2\sqrt{2}\,a$

(C) $\sqrt{\frac{5}{2}}\,a$ (D) $\frac{3}{\sqrt{2}}a$

38. If A and B are squares matrices such that $A^{2006} = 0$ and $AB = A + B$, then det(B) =

(A) 0

(B) 1

(C) 2

(D) 3

SECTION-2

One or More Than One Correct Type Questions

39. Let $\frac{dy}{dx} + y = f(x)$ where y is a continuous function of x with $y(0) = 1$ and $f(x) = \begin{cases} e^{-x} & \text{if } 0 \le x \le 2 \\ e^{-2} & \text{if } x > 2 \end{cases}$. Which of the following hold(s) good?

(A) $y(1) = 2e^{-1}$ (B) $y'(1) = -e^{-1}$

(C) $y(3) = -2e^{-3}$ (D) $y'(3) = -2e^{-3}$

40. Which of the following is (are) true?

(A) $\int_0^1 \sin(x^2 + 2x + 1)dx - \int_1^2 \sin(x^2)\,dx = 0$

(B) $\int_0^1 e^{\sin x}dx - 2\int_1^2 e^{\sin(2x-1)}dx = 0$

(C) $\int_{-5}^{-4} \sin(x^2 - 3)dx + \int_{-2}^{-1} \sin(x^2 + 12x + 33)\,dx = 0$

(D) $\int_{-4}^{4} \cos(x^2)dx - 8\int_0^1 \cos(16(2x-1)^2)\,dx = 0$

41. Given a function 'g' which has a derivative $g'(x)$ for every real x and satisfied $g'(0) = 2$ and g $(x + y) = e^y g(x) + e^x g(y)$ for all x and y then

(A) $\lim\limits_{x\to0}\frac{g(x)}{x} = 2$

(B) $g(x)$ is increasing function $\forall x \in R$

(C) $g(x)$ is one-one function $\forall x \in [-1,\infty)$

(D) Range of $g(x)$ is $\left[-\frac{2}{e}, \infty\right)$

SECTION-3

Non-Negative Integer Type Questions

42. If the equation $x^2 + px + q = 0$ ($p, q \in R$) and $x^3 + 3x^2 + 5x + 3 = 0$ have two roots common, then the value of $(p + q)$ is

43. If area between the curves $y = xe^x$ and $y = xe^{-x}$ and the line $x = 1$ is λ sq unit, then the value of $128\, e\lambda$ must be

44. A given quantity of metal is to be casted into solid half cylinder i.e., with a rectangular base and semicircular ends in order that the total surface area is minimum. The ratio of the height of the cylinder to the diameter of the semi-circular ends is $\pi : \pi + k$. Find k.

45. The distance of point of intersection of $\frac{x-4}{1} = \frac{y+3}{-4} = \frac{z+1}{7}$ and $\frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+10}{8}$ from (1, –4, 7) is λ. Then [λ] is equal to
(where [] represents g.i.f)

46. If the equation $4\sin\left(x+\frac{\pi}{3}\right)\cos\left(x-\frac{\pi}{6}\right) = a^2 + \sqrt{3}\sin 2x - \cos 2x$ has solution then number of the integral value(s) of a is

47. If $\alpha = \sum_{n=1}^{\infty} \tan^{-1}\left\{\frac{16(2n+1)}{(2n+1)^4 - 4(2n+1)^2 + 16}\right\}$ and $\sin\alpha = \frac{a}{b}$ where $a, b \in N$, then find the least value of $a^2 + b^2$.

SECTION-4

Paragraph Type Questions

Paragraph for question nos. (48- 49)

Let $N = \left(\sqrt[8]{5} + \sqrt[6]{2}\right)^{100}$

48. The number of irrational terms in the expansion of N is:

49. Sum of all possible values of r, such that T_{r+1} [$(r + 1)^{th}$ term] in the expansion of N is rational, is:

Paragraph for question nos. (50-51)

Eight players $P_1, P_2, \ldots P_8$ play a one on one knock out tournament, whenever player P_i plays with player P_j, player P_i will win if $i < j$. Assuming pairing of players is random in each round, then

50. The number of ways in which round 1 of the tournament can be planned is

51. The probability that P_4 reach the final is $\frac{p}{q}$ then minimum value of $(p + q)$ is (where $p, q \in Z$)



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if

[10]